Appendix A – Flashcards

Cover the right column and test yourself, or print and cut along the lines.

Energy carrier	A medium that stores/transports energy produced elsewhere (H2, electricity).
LHV of H2	120 MJ/kg = 33.3 kWh/kg (water as vapour).
HHV of H2	142 MJ/kg = 39.4 kWh/kg (water as liquid).
SMR reaction	CH4 + H2O → CO + 3H2 (endothermic).
Water-gas shift	CO + H2O → CO2 + H2 (exothermic).
POX reaction	CH4 + ½O2 → CO + 2H2 (exothermic).
Methane pyrolysis	CH4 → C(s) + 2H2; no CO2 formed.
Reversible cell voltage	1.23 V at 25 °C (ΔG).
Thermoneutral voltage	1.48 V at 25 °C (ΔH, HHV).
Charge per kg H2	≈ 26.6 kAh (Faraday's law, 2 e– per H2).
PEM anode reaction	2H2O → O2 + 4H+ + 4e–.
Alkaline cathode reaction	4H2O + 4e– → 2H2 + 4OH–.
SOEC electrolyte	Yttria-stabilised zirconia (YSZ), conducts O2–.
PEM catalysts	Anode IrOx; cathode Pt on carbon.
Alkaline electrolyte	KOH solution, 5–7 mol/L (IRENA).
Bipolar plate	Conducts current cell-to-cell, separates gases, carries flow channels.
PTL	Porous transport layer – water in, gas out, current through.
CCM	Catalyst-coated membrane – the PEM reaction heart.
PSA	Pressure swing adsorption – impurities adsorb at high P, desorb at low P.
TSA	Temperature swing adsorption dryer – molecular sieve regenerated by heat.
De-oxo	Catalytic removal of O2 by reaction with H2 to water.
SEC	Specific energy consumption, kWh per kg H2.
LCOH	Levelised cost of hydrogen, USD (or ₹) per kg.
Crossover	Gas permeating to the wrong electrode side – efficiency and safety issue.
Embrittlement	Loss of ductility of metals exposed to hydrogen.
Gas group IIC	Most easily ignited gas group (H2, acetylene).

Temperature class T1	Max equipment surface temp 450 °C – H2 rating.
Type IV tank	Polymer liner + full carbon-fibre wrap; 700 bar vehicles.
Salt cavern	Large-scale underground H2 storage.
Ortho–para conversion	Catalysed during liquefaction to avoid boil-off heat.
ISO 14687	Hydrogen fuel quality specification.
ISO 22734	Standard for water-electrolysis hydrogen generators.
DRI	Direct reduced iron – H2 reduces ore: $Fe_2O_3 + 3H_2 \rightarrow 2Fe + 3H_2O$.
Haber–Bosch	$N_2 + 3H_2 \rightleftharpoons 2NH_3$ – ammonia synthesis.

Appendix B – Knowledge Assessment

B1. Process sequencing

Put these green-hydrogen steps in order: (a) TSA drying (b) rectifier (c) de-oxo (d) electrolyser stack (e) compression (f) water treatment (g) gas–liquid separator (h) storage.

Put these SMR steps in order: (a) PSA (b) reformer (c) desulphurisation (d) LT shift (e) HT shift (f) heat recovery.

B2. Equipment identification

Description	Your answer
Vessel with demister that returns lye to the stack	
Twin molecular-sieve beds regenerated by heat	
Metal plate with flow channels that separates two cells	
Device converting AC to DC for the stack	
Catalyst bed that turns trace O2 into water	
Compressor with metal membrane isolating gas from oil	
Ceramic oxide-ion conductor operating near 800 °C	
Multi-bed adsorber unit burning its tail gas in the furnace	

B3. Chemical reactions – complete and balance

- $CH_4 + H_2O \rightarrow$ ____ + ____
- $CO + H_2O \rightarrow$ ____ + ____
- PEM anode: $2H_2O \rightarrow$ ____ + ____ + ____
- Fe_2O_3 + ___$H_2 \rightarrow$ ___Fe + ___H_2O
- N_2 + ___$H_2 \rightleftharpoons$ ___NH_3

B4. Calculation questions

- A 5 MW PEM plant runs at 55 kWh/kg. How many kg H_2 per hour, and how many tonnes of O_2 per day at full load?
- A stack operates at 1.9 V/cell. Estimate stack-level energy per kg H_2.
- Electricity costs 40 USD/MWh and SEC is 52 kWh/kg. What is the electricity cost per kg?

B5. Industrial case studies

Refinery switch to green H2
A refinery uses 30 t/day of grey hydrogen from SMR. Management wants to replace 20 % with green hydrogen. List the plant units needed, estimate the electrolyser size at 52 kWh/kg running 24 h, and name two integration risks.

Alkaline vs PEM selection
A solar-only site (no storage) will feed an electrolyser. Which technology would you shortlist and why? Consider dynamic response, minimum load, footprint, materials and cost (use Module 4 table).

Refuelling station supply
A bus depot needs fuel-cell-grade hydrogen at 350 bar. Choose between on-site electrolysis and tube-trailer delivery; list purification, compression and safety requirements.

Answer key

- B1 green: f → b → d → g → c → a → e → h. SMR: c → b → f → e → d → a.
- B2: separator; TSA dryer; bipolar plate; transformer-rectifier; de-oxo; diaphragm compressor; SOEC (YSZ) cell; PSA.
- B3: $CO + 3H_2$; $CO_2 + H_2$; $O_2 + 4H^+ + 4e^-$; 3, 2, 3; 3, 2.
- B4: 5,000 ÷ 55 = 90.9 kg/h; $O_2 = 8 \times 90.9 \times 24 \div 1{,}000 = 17.5$ t/day. 26.6 kAh × 1.9 V ≈ 50.5 kWh/kg. 52 × 40 ÷ 1,000 = 2.08 USD/kg.
- B5 (refinery): 6 t/day = 250 kg/h × 52 = 13 MW electrolyser (24 h) + water treatment, purification, compression to the hydrogen header pressure, buffer storage; risks: intermittent supply vs steady refinery demand, purity/pressure matching, power-supply reliability.
- B5 (solar site): PEM usually shortlisted for fast dynamics and wide load range; alkaline for lower capex if a battery/grid smooths supply.
- B5 (bus depot): ISO 14687 quality, drying + de-oxo (electrolysis) or quality testing (trailer), multi-stage compression to above 350 bar for cascade storage, ISO 19880-1 station design, gas & flame detection, area classification.

Appendix C – Glossary (alphabetical, searchable in any PDF reader with Ctrl+F)

Term	Meaning
AEL	Alkaline electrolysis
AEM	Anion exchange membrane electrolysis
ATR	Autothermal reforming
BoP	Balance of plant – everything except the stack
CCS	Carbon capture and storage
CCM	Catalyst-coated membrane
CRF	Capital recovery factor
DCS	Distributed control system
DRI	Direct reduced iron
EDI	Electro-deionisation
ESD	Emergency shutdown
HHV / LHV	Higher / lower heating value
HRS	Hydrogen refuelling station
LCOH	Levelised cost of hydrogen
LFL / LEL	Lower flammable / explosive limit
LH2	Liquid hydrogen
LOHC	Liquid organic hydrogen carrier
NGHM	India's National Green Hydrogen Mission
PEM	Proton exchange (polymer electrolyte) membrane
PFSA	Perfluorosulfonic acid membrane (e.g. Nafion)
POX	Partial oxidation
PSA	Pressure swing adsorption
PTL	Porous transport layer
RO	Reverse osmosis
SEC	Specific energy consumption (kWh/kg)
SIS	Safety instrumented system
SMR	Steam methane reforming
SOEC	Solid oxide electrolysis cell
TSA	Temperature swing adsorption
WGS	Water-gas shift
YSZ	Yttria-stabilised zirconia

Appendix D – Learning Progress Tracker

Tick when completed, write your quiz score (out of 5), and list topics to revise.

Module	Done ☐	Date	Quiz score /5	Topics mastered	Topics to revise
1 Introduction	☐				
2 Production technologies	☐				
3 Detailed process	☐				
4 Electrolyser deep dive	☐				
5 Plant layout & equipment	☐				
6 Process simulation	☐				
7 Purification & conditioning	☐				
8 Storage & transport	☐				
9 Applications	☐				
10 Safety & standards	☐				
Appendix B assessment	☐				

Mastery rule
5/5 = mastered · 4/5 = revise flashcards for the missed item · 3/5 or less = re-read the module and redo the quiz after 2 days.

Appendix E – References

Ref	Source	Link
[R1]	IRENA (2020). Green Hydrogen Cost Reduction: Scaling up Electrolysers to Meet the 1.5 °C Climate Goal. Table 1, Table ES1, pp. on water use and compression.	https://www.irena.org/-/media/Files/IRENA/Agency/Publication/2020/Dec/IRENA_Green_hydrogen_cost_2020.pdf
[R2]	U.S. Department of Energy, Hydrogen and Fuel Cell Technologies Office. Hydrogen Production: Natural Gas Reforming.	https://www.energy.gov/eere/fuelcells/hydrogen-production-natural-gas-reforming
[R3]	IEA (2024). Global Hydrogen Review 2024 – Hydrogen demand.	https://www.iea.org/reports/global-hydrogen-review-2024/hydrogen-demand
[R4]	IEA (2025). Global Hydrogen Review 2025 – Demand.	https://www.iea.org/reports/global-hydrogen-review-2025/demand
[R5]	IEA (2023). Towards Hydrogen Definitions Based on their Emissions Intensity – Executive summary.	https://www.iea.org/reports/towards-hydrogen-definitions-based-on-their-emissions-intensity/executive-summary
[R6]	Stolten, D. (ed.). Hydrogen and Fuel Cells: Fundamentals, Technologies and Applications, Wiley-VCH, Chapter 1 (properties).	https://application.wiley-vch.de/books/sample/3527333428_c01.pdf
[R7]	NREL (2019). Manufacturing Cost Analysis for PEM and Alkaline Water Electrolyzers (technology characteristics).	https://www.nrel.gov/docs/fy19osti/70380.pdf
[R8]	Renewable Watch (2025). Global Hydrogen Review 2025: IEA (demand ~100 Mt in 2024).	https://renewablewatch.in/2025/09/21/global-hydrogen-review-2025-iea/
[R9]	MNRE / PIB. National Green Hydrogen Mission; pilot projects for hydrogen buses and trucks (3 Mar 2025).	https://nghm.mnre.gov.in/admin/uploads/174110589023911Press%20Release_PIB_H2_fuelled_buses_trucks.pdf
[R10]	MNRE (2023). Green Hydrogen Standard for India (≤2 kg CO2-eq/kg H2, well-to-gate).	https://mnre.gov.in/

Standard reaction enthalpies, Faraday constant, cell voltages and ideal-gas compression are textbook physical chemistry values. Process descriptions without a numeric reference are general engineering knowledge and are intentionally kept qualitative.